Multi-View Semantic 3D Gaussian Splatting for Component-Aware Bridge Reconstruction from UAV Imagery

Rendering bridge appearance and structural components from arbitrary viewpoints

Quoc-Bao Ta[1] Trung Ha Nguyen[2] Thai Ha Dang[3]

[1]Department of Civil Engineering, University of Texas Rio Grande Valley, Edinburg, TX, USA
[2]Jisan Construction Co., Ltd., Busan, South Korea
[3]Department of Electrical Engineering, University of North Texas, Denton, TX, USA

IC-SHM Competition – Project 2
September 2026

Competition task

- IC-SHM 2026 Project 2 asks for a renderable semantic reconstruction of one suspension bridge from posed UAV imagery.
- A query camera pose must produce an RGB image and a five-class map: background, deck, stay cable, tower, and foundation.
- The organizers use PSNR, SSIM, LPIPS, and semantic mIoU on an unreleased blind set. Our results use a separate local 30-view holdout [1].
- The target is a component-aware scene representation. Damage detection, deformation measurement, and survey-grade geometry are outside this evaluation.

Objective

Learn one 3D representation that renders aligned appearance and component labels from supplied or constructed viewpoints.

Released data and required output

- 400 UAV photographs at 1320×989 pixels.
- 300 images contain polygons for four bridge components and background.
- 100 images contain RGB only. Our pipeline generates pseudo-labels so these views can provide semantic supervision.
- SIMPLE_RADIAL calibration, camera poses, and COLMAP feature tracks are supplied, but no precomputed 3D points.

Renderer interface

Input: rotation, translation, intrinsics, image size.

Output: RGB $\hat{I}$ and class map $\hat{M} \in \{0, \ldots, 4\}^{H \times W}$.

Local partition

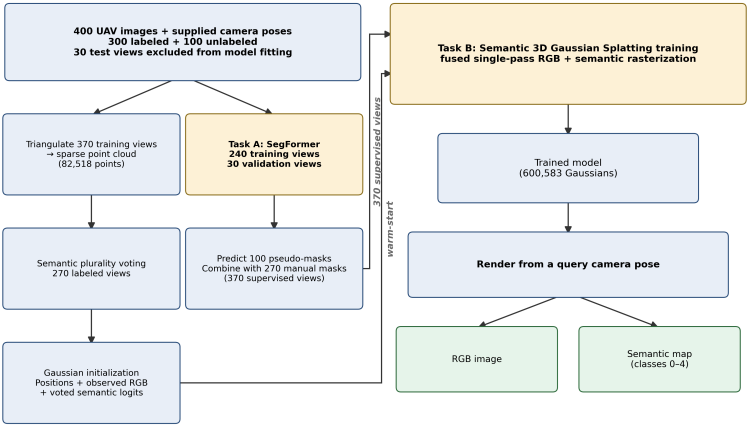
240 labeled train + 30 labeled validation + 30 labeled test; 100 unlabeled views remain separate.

Related methods and our design choice

Family	Capability	Choice for this task
NeRF	Implicit novel-view synthesis	Useful precedent, but no explicit primitive for a bridge-component label.
3D Gaussian Splatting [3]	Explicit primitives and differentiable rasterization	Store RGB and five semantic logits on every Gaussian.
Semantic/feature scene models [4,5]	Render labels or learned features	Optimize fixed-class logits directly instead of open-vocabulary features.
SegFormer [6]	Efficient multi-scale segmentation	Generate pseudo-labels for the 100 unlabeled views.
COLMAP geometry	Posed multi-view tracks	Triangulate after excluding test observations.

The contribution is a task-specific integration for UAV bridge data, not a claim that semantic attribute rendering is new.

Proposed pipeline



- **Task A** trains on 240 views, selects on 30, and labels 100 unlabeled views.
- **Task B** jointly optimizes RGB and semantics from 370 supervised views.
- One trained scene renders an RGB image and a five-class map from a query pose.

Data split and test isolation

- Every tenth labeled image forms the 30-view test split.
- Every ninth image among the remaining 270 forms Task A's 30-view validation split; 240 images train SegFormer.
- Task B uses 270 non-test labeled views and 100 pseudo-labeled views.
- Test images contribute to no Task A training or selection and no Task B model-building step.

Excluded test information

Task B losses
semantic voting
sparse triangulation
observed-color initialization

Scope

Organizer-supplied calibration and poses are retained, so evaluation conditions on supplied camera geometry.

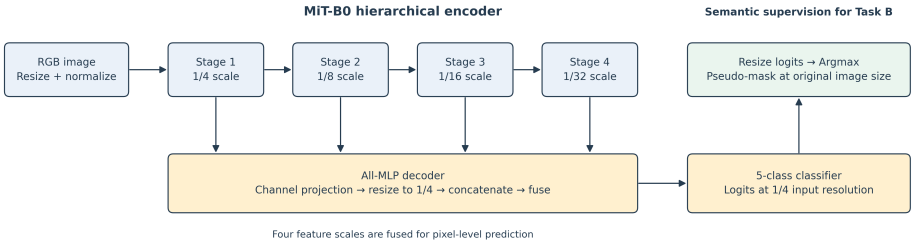
Camera geometry and sparse initialization

- Removing test observations leaves 86,319 tracks from the 370-image training pool.
- LO-RANSAC uses a 2° angular threshold, a 0.5° minimum triangulation angle, and cheirality checks.
- Triangulation produces 84,086 points with 0.49-pixel mean reprojection error; a quartile-distance filter leaves 82,518 initial Gaussians.
- Track-coordinate color sampling and semantic voting use original images. Task B images and masks are undistorted into the same pinhole coordinates.
- Nearest-neighbor mask remapping preserves discrete class IDs.

Coordinate rule

Original space: feature tracks, sampled colors, votes. Undistorted space: Task B rendering, losses, and evaluation.

Task A: SegFormer pseudo-labeling



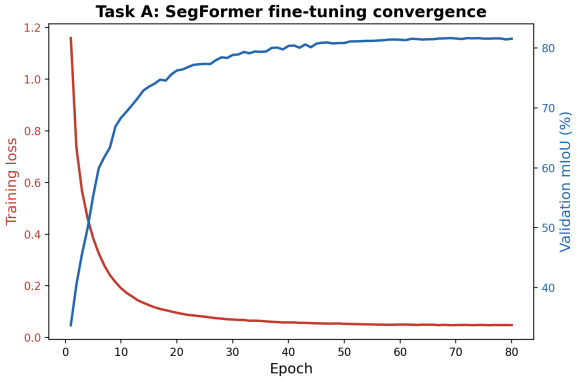
- The MiT-B0 encoder extracts four feature scales; the all-MLP decoder fuses them and predicts five-class logits [6].
- Training uses 240 labeled images and checkpoint selection uses 30 validation views. The 30 test images never influence Task A.
- Resizing logits to the original image dimensions followed by argmax generates masks for the 100 unlabeled images.

Task A protocol and validation result

Class	IoU
deck	92.80%
stay_cable	89.17%
tower	78.20%
foundation	66.48%
background	98.64%
Structural mIoU	**81.67%**

80

epochs, batch 8, AdamW, 512 × 384; best checkpoint at epoch 69.



Metric boundary

Task A uses its 30-image internal validation split. These values are not Task B test results.

Semantic Gaussian representation

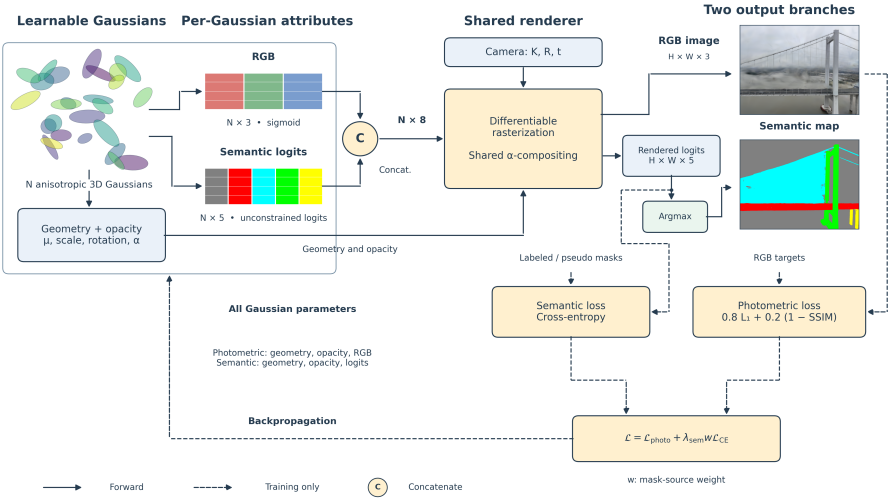
$$g_k = \{\mu_k, s_k, q_k, o_k, c_k, \ell_k\}, \qquad c_k \in (0,1)^3, \quad \ell_k \in \mathbb{R}^5.$$

- μ_k and s_k describe position and scale; q_k is a unit rotation quaternion and o_k is opacity.
- Geometry, opacity, RGB, and semantic logits remain jointly trainable; RGB is view-independent.
- Position and observed RGB originate from one sparse point, while multi-view votes initialize ℓ_k.

Eight trainable rendering channels

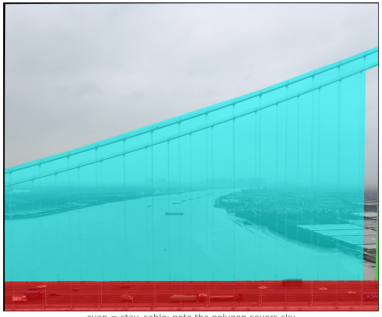
The renderer concatenates three RGB values c_k and five class logits ℓ_k for each Gaussian.

Semantic Gaussian architecture



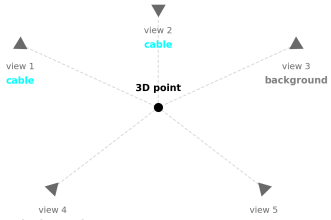
Semantic warm-start

Real example: GT mask overlaid on an undistorted UAV image



cyan = stay_cable; note the polygon covers sky
and river far beyond the cable strands

Illustrative example: multi-view vote for one 3D point



cable (2) ties background (2), deck (1) trails
→ plurality tie broken by fixed priority (thin/rare structures first)
→ point labeled cable

- Each labeled observation votes at its track coordinate. The plurality winner initializes $+2$ for its class and -2 for the others.
- Ties use the priority cable, tower, foundation, deck, background. Pseudo-labels supervise training but do not vote during initialization.
- The broad cable polygon defines the semantic target; it does not resolve individual strands.

Fused RGB and semantic rendering

$$T_{kp} = \prod_{j<k}(1 - a_{jp}), \qquad T_{\mathrm{end},p} = \prod_{k}(1 - a_{kp}).$$

$$\hat{I}_p = \sum_k T_{kp} a_{kp} c_k + T_{\mathrm{end},p} b_I, \qquad z_p = \sum_k T_{kp} a_{kp} \ell_k + T_{\mathrm{end},p} b_\ell.$$

$$\hat{M}_p = \arg\max_c z_{pc}, \qquad b_I = (.5, .5, .5), \quad b_\ell = (4, -4, -4, -4, -4).$$

Implementation consequence

Three RGB channels and five logits form one eight-channel gsplat pass [7]. Both outputs share projection, visibility, depth order, and alpha-compositing weights.

Joint optimization objective

$$\frac{\partial \mathcal{L}_{\mathrm{CE},p}}{\partial \ell_k} = T_{kp} a_{kp} \big(\mathrm{softmax}(z_p) - e_{y_p}\big).$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{photo}} + \lambda_{\mathrm{sem}} \omega \mathcal{L}_{\mathrm{sem}}, \qquad \mathcal{L}_{\mathrm{photo}} = .8\mathcal{L}_1 + .2(1 - \mathrm{SSIM}).$$

$$\lambda_{\mathrm{sem}} = .5, \qquad \omega = \begin{cases} 1 & \text{manual annotation,} \\ .5 & \text{pseudo-label.} \end{cases}$$

- Semantic loss updates logits and shared geometry/opacity through compositing weights.
- Photometric loss directly updates RGB, geometry, and opacity, with no direct gradient to logits.

Mask alignment and observed-color initialization

Coordinate alignment

- Polygons and SegFormer predictions originate in distorted photographs.
- Task B renders undistorted pinhole images.
- Images and masks receive the same remap; masks use nearest-neighbor interpolation.
- Voting remains in original coordinates because COLMAP tracks reference original images.

Appearance initialization

- Sample each point's RGB at every retained non-test observation.
- Average observations to initialize one Gaussian's color.
- Unlabeled views may contribute RGB; test views remain excluded.
- Scales use three-neighbor distance; rotation starts at identity and opacity at 0.3.

Task B experimental setup

- 370 supervised views: 270 manual masks and 100 pseudo-masks.
- 40,000 iterations at 1320×989; seed 42.
- Adam rates: means 1.6×10^{-4}, scales 5×10^{-3}, rotation 10^{-3}, opacity 5×10^{-2}, RGB/logits 2.5×10^{-3}.
- Density control every 100 steps from 500 to 15,000; 600,000-Gaussian soft cap.

Representation growth

82,518 initial points
600,583 final Gaussians
stable after step 7,400

Compute

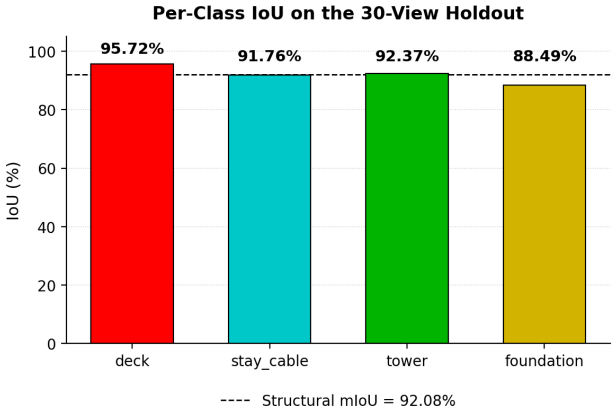
NVIDIA RTX 3080 (10 GB)
about 46 minutes at full resolution

Task B aggregate test results

Metric	Value
PSNR	22.43 dB
SSIM	0.854
LPIPS	0.321
Structural mIoU	**92.08%**
Five-class mIoU	93.5%
Illustrative Accuracy Score	0.823

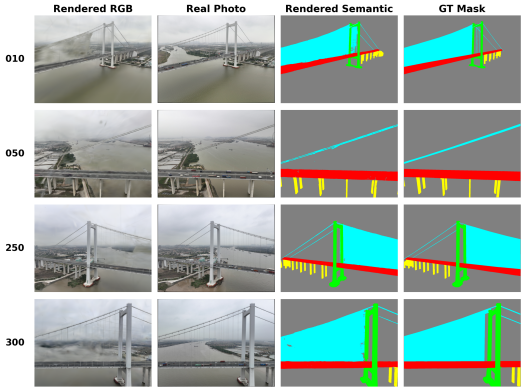
- Metrics use 30 held-out, undistorted views.
- Structural mIoU averages four bridge classes equally; background is excluded.
- The illustrative Accuracy Score is our interpretation of the brief, not a blind-test score.
- The split evaluates interpolation near the released trajectory with supplied geometry.

Task B per-class semantic results



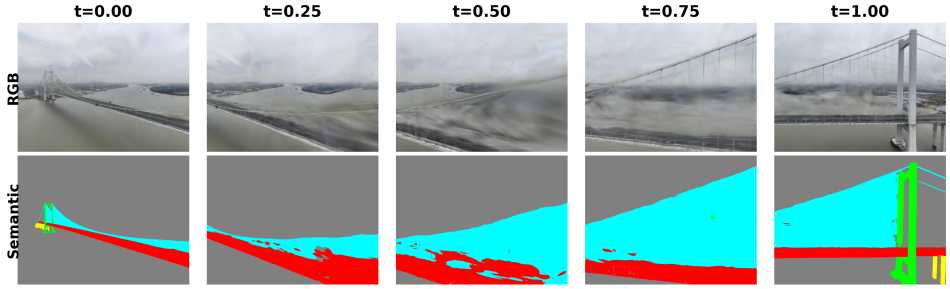
- Deck: 95.72%.
- Tower: 92.37%.
- Stay cable: 91.76%.
- Foundation: 88.49%.
- Background: 99.20%, reported separately.

Qualitative held-out views



- View 250 is the cleanest RGB example; views 010 and 300 show visible blur and streaking.
- Semantic regions remain recognizable despite appearance artifacts, but four examples do not replace full-holdout statistics.

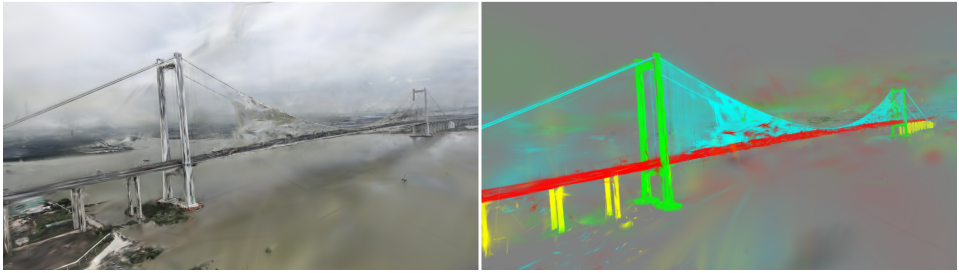
Rendering between observed viewpoints



- Rotation uses quaternion SLERP; translation interpolates world-to-camera translation between test poses 280 and 300.
- Intermediate poses have no ground-truth photographs and provide a qualitative interface demonstration only.
- RGB blur and semantic holes increase along parts of the path. Shared rasterization preserves alignment, not correctness.

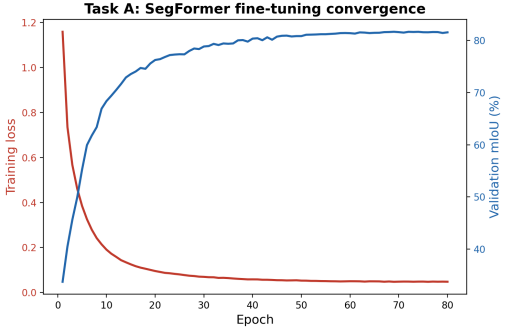
Explicit RGB and semantic splats

(a) True-color splat render **(b) Colored by predicted semantic class**

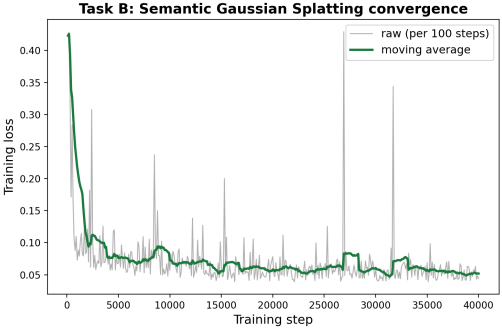


- One trained representation appears as learned RGB and per-Gaussian argmax class colors.
- Both screenshots use the same viewer camera pose, enabling direct spatial comparison of RGB appearance and per-Gaussian semantic colors.
- Per-Gaussian recoloring differs from evaluated semantic rendering, which composites logits before argmax.

Training convergence



Task A: best validation mIoU 81.67% at epoch 69.



Task B: sampled-view loss and moving average.

- Task A curves summarize epoch averages; Task B logs one sampled view every 100 steps.
- Their loss scales are not comparable. Final Task B performance is measured on the test split.

Resolution ablation

Training	PSNR	SSIM	LPIPS	$\mathrm{mIoU}_{\mathrm{struct}}$
Half	22.03	.838	.330	89.71%
Full	**22.43**	**.854**	**.321**	**92.08%**

- Matched 40,000-step runs share the split, poses, initialization, losses, density schedule, and full-resolution evaluation.
- Full resolution improves PSNR by 0.40 dB, SSIM by 0.016, and structural mIoU by 2.37 points; LPIPS decreases by 0.009.
- Training time increases from about 18 to 46 minutes on the same RTX 3080.
- One run per resolution does not quantify random-seed uncertainty.

What the evaluation establishes

- One explicit Gaussian scene renders aligned RGB and five-class semantic outputs.
- On 30 local test views, the final model reaches 22.43 dB PSNR and 92.08% structural mIoU.
- Test observations are excluded from losses, voting, triangulation, and color initialization.
- The interleaved split samples viewpoints near training frames and mainly tests interpolation under supplied geometry.
- Full-resolution training outperforms the matched half-resolution run within the tested budget.

Claim boundary

The results establish agreement with released images and masks near the flight trajectory, not unrestricted viewpoint generalization or metric 3D accuracy.

Limitations

- The interleaved holdout does not test a spatially separated trajectory segment.
- Supplied poses and calibration originate from the released collection; sensitivity to independent pose estimation is unknown.
- Coarse cable polygons cannot validate individual strand geometry or boundaries.
- Constructed novel poses show artifacts in both RGB and semantics and have no ground truth.
- The system has not been evaluated for damage detection, deformation, cable force, or structural response.

Conclusion

- SegFormer expands semantic supervision from 270 manually labeled views to 370 views.
- Sparse geometry, observed colors, and plurality votes initialize an explicit semantic Gaussian scene.
- One differentiable rasterization pass produces aligned RGB and five semantic-logit channels.
- Local test performance is 22.43 dB PSNR, 0.854 SSIM, 0.321 LPIPS, and 92.08% structural mIoU.
- Full-resolution training improves every reported metric relative to the matched half-resolution run.
- Future tests should use spatially separated views, independent geometry, and finer cable annotations.

Outcome

A renderable, component-labeled bridge representation with explicit limits on geometric and out-of-trajectory claims.

Selected references

1. IC-SHM 2026 Organizing Committee, "The 4th International Project Competition for Structural Health Monitoring," Project 2 competition brief, pp. 9–10, 2026.
2. J. L. Schönberger and J.-M. Frahm, "Structure-from-Motion Revisited," CVPR, 2016.
3. B. Kerbl et al., "3D Gaussian Splatting for Real-Time Radiance Field Rendering," ACM Transactions on Graphics, 42(4), Article 139, 2023.
4. S. Zhi et al., "In-Place Scene Labelling and Understanding with Implicit Scene Representation," ICCV, 2021.
5. S. Zhou et al., "Feature 3DGS: Supercharging 3D Gaussian Splatting to Enable Distilled Feature Fields," CVPR, 2024.
6. E. Xie et al., "SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers," NeurIPS, 2021.
7. V. Ye et al., "gsplat: An Open-Source Library for Gaussian Splatting," Journal of Machine Learning Research, 26(34), 1–17, 2025.